

QuantEdge Complete Application Flow & Market Session User Flow

1. Core Product Behavior

QuantEdge behaves like a realtime trading operating system throughout the entire market session. The platform continuously watches the market, detects opportunities, projects movement, monitors positions, suggests exits, validates outcomes, and learns from results.

2. Daily Market Lifecycle

The platform lifecycle follows the Indian stock market timing:

- Pre-Market Phase (9:00 AM – 9:15 AM)
 - Active Market Phase (9:15 AM – 3:30 PM)
 - Post-Market Analysis Phase (After 3:30 PM)
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3. Application Startup Flow

The application opens directly into the realtime trading dashboard without authentication, onboarding, or user management. The system immediately initializes market connections, websocket streams, forecasting engines, indicator calculations, and realtime analytics.

4. Initial Dashboard State

Before market opens, the dashboard displays:

- Market status and countdown
- NIFTY and BANKNIFTY movement
- Volatility indicators
- Prediction engine initialization states
- Scanner warmup status

The system waits for enough live market data before generating signals.

5. Market Opening Flow

At market open, live candles begin forming and option chains start updating. The system immediately starts realtime calculations including momentum analysis, volatility tracking, trend evaluation, and signal generation.

6. Initial Volatility Protection

During the first 5–15 minutes after market open, the platform reduces prediction confidence and avoids aggressive recommendations due to high opening volatility and fake directional moves.

7. Live Market Scanner Flow

The market scanner continuously evaluates:

- Volume spikes
- Momentum acceleration
- Trend strength
- Volatility expansion
- Breakout probability
- Sector strength
- Option activity

The strongest opportunities are ranked dynamically in realtime.

8. Opportunity Analysis Flow

When a recommendation is opened, the system displays:

- Market direction
 - Confidence score
 - Expected movement range
 - Risk level
 - Suggested entry zone
 - Suggested exit zone
 - Stop loss
 - Analytical reasoning behind the signal
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9. Investment Entry Flow

The user manually inputs:

- Invested amount
- Entry price
- Quantity or lot count

After submission, the platform activates realtime position intelligence mode for that specific trade.

10. Active Position Intelligence

The platform continuously analyzes:

- Momentum strength

- Candle behavior
- Volume pressure
- Volatility
- Trend exhaustion
- Reversal probability

The system projects short-term movement ranges for the next 1 minute, 5 minutes, and 10 minutes.

11. Projection Engine Workflow

Multiple prediction engines independently generate projected future movement paths:

- Momentum Engine
- Volatility Engine
- Mean Reversion Engine
- ML Engine
- Monte Carlo Engine

Each engine produces projected candle paths, confidence scores, and movement ranges.

12. Candle Projection Visualization

Projected candles appear directly on the chart using semi-transparent and dashed visual styles. Each prediction engine renders its own projection path and probability zone.

13. Realtime Monitoring Flow

The system continuously recalculates:

- Momentum
- Volatility
- Trend pressure
- Reversal probability
- Projection deviation
- Volume behavior

Confidence levels update dynamically as the market evolves.

14. Exit Intelligence Flow

The Exit Intelligence Engine determines:

- Whether momentum is weakening
- Whether reversal probability is increasing
- Whether profit booking is likely

The platform dynamically suggests:

- Hold
 - Partial Exit
 - Aggressive Exit
 - Full Sell
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15. Trade Closure Flow

Once the trade is exited, the platform stores:

- Entry data
 - Exit data
 - Projection history
 - Accuracy metrics
 - Actual market behavior
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16. Post-Market Validation

After market close, the platform compares:

- Predicted movement
- Actual movement
- Predicted ranges
- Actual ranges
- Suggested exits
- Real outcomes

The system calculates model accuracy and forecasting performance.

17. Daily Accuracy Dashboard

The system generates daily analytics showing:

- Engine accuracy rankings
 - Best performing models
 - Weakest performing models
 - Regime-based performance
 - Projection reliability
 - Forecast deviation statistics
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18. Adaptive Intelligence

Over time, the platform learns which prediction engines perform best under specific market conditions. Confidence weighting dynamically adapts based on historical accuracy and current market behavior.

19. Market Sleep Mode

After post-market analysis completes, the platform enters market sleep mode and displays finalized daily analytics, prediction reports, and readiness for the next trading session.

20. Final Product Experience

The complete experience should feel like a calm, intelligent, realtime market operating system that continuously helps determine:

- What to trade
 - Why to trade
 - How risky it is
 - What may happen next
 - When to exit intelligently
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